JAVA

PART - A

1. Program to accept student name and marks in three subjects. Find the total marks, average and grade (depending on the average marks).

Code:

```
import java.io.*;
public class Grade {
   public static void main(String[] args) throws IOException {
      String name, g;
      float m1, m2, m3, total, average;
      DataInputStream dis = new DataInputStream(System.in);
      System.out.print("Enter student name: ");
      name = dis.readLine();
      System.out.println("Enter marks in 3 subjects: ");
      m1 = Float.parseFloat(dis.readLine());
      m2 = Float.parseFloat(dis.readLine());
      m3 = Float.parseFloat(dis.readLine());
      System.out.println("\nStudent name: " + name);
      System.out.println("Marks in subject 1: " + m1);
      System.out.println("Marks in subject 2: " + m2);
      System.out.println("Marks in subject 3: " + m3);
      total = m1 + m2 + m3;
      System.out.println("Total = " + total);
      average = total / 3;
      System.out.println("Average = " + average);
      if (m1 <= 35 || m2 <= 35 || m3 <= 35)
         g = "Fail";
      else if (average >= 85)
```

```
            g = "A";
        else if (average >= 70)
            g = "B";
        else if (average >= 60)
            g = "C";
        else
            g = "D";
        System.out.println("Grade = " + g);
    }
}
```

Output:

```
Microsoft Windows [Version 10.0.26200.8037]
(c) Microsoft Corporation. All rights reserved.

D:\Documents\Java\LAB PROGRAMS>javac Grade.java
Note: Grade.java uses or overrides a deprecated API.
Note: Recompile with -Xlint:deprecation for details.

D:\Documents\Java\LAB PROGRAMS>java Grade
Enter student name: Ashwith Frank
Enter marks in 3 subjects:
94
89
97

Student name: Ashwith Frank
Marks in subject 1: 94.0
Marks in subject 2: 89.0
Marks in subject 3: 97.0
Total = 280.0
Average = 93.333336
Grade = A
```

2. Program, which reads two numbers having same number of digits. The program outputs the sum of product of corresponding digits.(Hint Input 327 and 539 output 3x5+2x3+7x9=84)

Code:

```
import java.io.*;

public class SumOfProduct {

    public static void main(String[] args) throws IOException {

        int result = 0;

        DataInputStream dis = new DataInputStream(System.in);

        System.out.println("Enter two numbers of same digits:");

        int num1 = Integer.parseInt(dis.readLine());

        int num2 = Integer.parseInt(dis.readLine());

        while (num1 > 0 && num2 > 0) {

            result = result + ((num1 % 10) * (num2 % 10));

            num1 /= 10;

            num2 /= 10;

        }

        System.out.println("Sum of product of corresponding digits: " + result);

    }

}
```

Output:

```
Microsoft Windows [Version 10.0.26200.8037]
(c) Microsoft Corporation. All rights reserved.

D:\Documents\Java\LAB PROGRAMS>javac SumOfProduct.java
Note: SumOfProduct.java uses or overrides a deprecated API.
Note: Recompile with -Xlint:deprecation for details.

D:\Documents\Java\LAB PROGRAMS>java SumOfProduct
Enter two numbers of same digits:
327
539
Sum of product of corresponding digits: 84
```

3. Program to input Start and End limits and print all Fibonacci numbers between the ranges. (Use for loop)

Code:

```
import java.util.Scanner;
class Fibonacci {
    public static void main(String args[]) {
        int l, u, f1 = 0, f2 = 1, f3;
        Scanner s = new Scanner(System.in);
        System.out.println("Enter the lower and upper limit:");
        l = s.nextInt();
        u = s.nextInt();
        System.out.println("Fibonacci numbers in the given range: ");
        while (f1 <= u) {
            if (f1 >= l && f1 <= u)
                System.out.print(f1 + " ");
            f3 = f1 + f2;
            f1 = f2;
            f2 = f3;
        }
    }
}
```

Output:

```
Microsoft Windows [Version 10.0.26200.8037]
(c) Microsoft Corporation. All rights reserved.

D:\Documents\Java\LAB PROGRAMS>javac Fibonacci.java

D:\Documents\Java\LAB PROGRAMS>java Fibonacci
Enter the lower and upper limit:
0 20
Fibonacci numbers in the given range:
0 1 1 2 3 5 8 13
D:\Documents\Java\LAB PROGRAMS>
```

4. Define a class named Pay with data members String name, double salary, double da, double hra, double pf, double grossSal, double netSal and methods: Pay(String n, double s) - Parameterized constructor to initialize the data members, void calculate() - to calculate the following salary components, and void display() - to display the employee name, salary and all salary components.

- **Dearness Allowance = 15% of salary**
- **House Rent Allowance = 10% of salary**
- **Provident Fund = 12% of salary**
- **Gross Salary = Salary + Dearness Allowance + House Rent Allowance**
- **Net Salary = Gross Salary - Provident Fund**

Write a main method to create object of the class and call the methods to compute and display the salary details. [class basics]

Code:

```
import java.io.*;

public class Pay {

    String name;

    double salary;

    double da;

    double hra;

    double pf;

    double grossSal;

    double netSal;


    public Pay(String n, double s) {

        name = n;

        salary = s;

        da = 0;

        hra = 0;
```

```
    pf = 0;
    grossSal = 0;
    netSal = 0;
}

void calculate() {
    da = salary * 15.0 / 100;
    hra = salary * 10.0 / 100;
    pf = salary * 12.0 / 100;
    grossSal = salary + da + hra;
    netSal = grossSal - pf;
}

void display() {
    System.out.println("\nEmployee Name : " + name);
    System.out.println("Salary : " + salary);
    System.out.println("Dearness Allowance : " + da);
    System.out.println("House Rent Allowance : " + hra);
    System.out.println("Provident Fund: " + pf);
    System.out.println("Gross Salary : " + grossSal);
    System.out.println("Net Salary : " + netSal);
}

public static void main(String args[]) throws IOException {
    DataInputStream dis = new DataInputStream(System.in);
    System.out.print("Enter Employee Name: ");
```

```
        String empName = dis.readLine();
        System.out.print("Enter Salary: ");
        double empSal = Double.parseDouble(dis.readLine());
        Pay obj = new Pay(empName, empSal);
        obj.calculate();
        obj.display();
    }
}
```

Output:

```
Microsoft Windows [Version 10.0.26200.8037]
(c) Microsoft Corporation. All rights reserved.

D:\Documents\Java\LAB PROGRAMS>javac Pay.java
Note: Pay.java uses or overrides a deprecated API.
Note: Recompile with -Xlint:deprecation for details.

D:\Documents\Java\LAB PROGRAMS>java Pay
Enter Employee Name: Ashwith Frank
Enter Salary: 25000

Employee Name : Ashwith Frank
Salary : 25000.0
Dearness Allowance : 3750.0
House Rent Allowance : 2500.0
Provident Fund: 3000.0
Gross Salary : 31250.0
Net Salary : 28250.0
```

5. Program to create a class DISTANCE with the data members feet and inches. Use a constructor to read the data and a member function Sum ( ) to add two distances by using objects as method arguments and show the result. (Input and output of inches should be less than 12.).

Code:

```
import java.util.*;

class Distance {

    int feet;

    int inches;


    public Distance() {

        feet=0;

        inches=0;

    }


    public void getDistance() {

        Scanner sc = new Scanner(System.in);

        System.out.print("Enter feet: ");

        feet = sc.nextInt();

        System.out.print("Enter inches: ");

        inches = sc.nextInt();

        System.out.println();

    }


    public void showDistance() {

        System.out.println("Feet: " + feet + "\nInches: " + inches);

    }
```

```
    public void Sum(Distance D1, Distance D2) {
        inches = D1.inches + D2.inches;
        feet = D1.feet + D2.feet;
    }
}

public class AddTwoDistances {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        Distance D1 = new Distance();
        Distance D2 = new Distance();
        Distance D3 = new Distance();
        System.out.println("Enter first distance: ");
        D1.getDistance();
        System.out.println("Enter second distance: ");
        D2.getDistance();
        D3.Sum(D1, D2);
        if (D3.inches > 12) {
            D3.feet++;
            D3.inches = D3.inches - 12;
        }
        System.out.println("Total distance is: ");
        D3.showDistance();
    }
}
```

Output:

```
Microsoft Windows [Version 10.0.26200.8037]
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D:\Documents\Java\LAB PROGRAMS>javac AddTwoDistances.java

D:\Documents\Java\LAB PROGRAMS>java AddTwoDistances
Enter first distance:
Enter feet: 15
Enter inches: 4

Enter second distance:
Enter feet: 21
Enter inches: 5

Total distance is:
Feet: 36
Inches: 9
```

6. Program to create a class "Matrix" that would contain integer values having varied numbers of columns for each row. Print row-wise sum.

Code:

```
import java.util.Scanner;
public class Matrix {
    int m, n;
    int a[][];

    Matrix(short rows) {
        m = rows;
        n = 0;
        a = new int[m][];

    }

    void read() {
        int i, j;
        Scanner sc = new Scanner(System.in);
        for (i = 0; i < m; i++) {
            System.out.println("Enter the number of coloumns in row " + (i + 1));
            short n = sc.nextShort();
            a[i] = new int[n];
        }
        for (i = 0; i < m; i++) {
            System.out.println("Enter the elements of row " + (i + 1));
            for (j = 0; j < a[i].length; j++) {
```

```
            a[i][j] = sc.nextInt();
        }
    }
}

void display() {
    int i, j;
    System.out.println("\nThe given Matrix");
    for (i = 0; i < m; i++) {
        for (j = 0; j < a[i].length; j++)
            System.out.print(a[i][j] + " ");
        System.out.println();
    }
    System.out.println();
    for (i = 0; i < m; i++) {
        int sum = 0;
        for (j = 0; j < a[i].length; j++)
            sum = sum + a[i][j];
        System.out.println("Sum of row " + (i + 1) + " = " + sum);
    }
}

public static void main(String args[]) {
    short m;
    Scanner sc = new Scanner(System.in);
    System.out.print("Enter the number of rows: ");
```

```
        m = sc.nextShort();
        Matrix obj = new Matrix(m);
        obj.read();
        obj.display();
    }
}
```

Output:

```
Microsoft Windows [Version 10.0.26200.8037]
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D:\Documents\Java\LAB PROGRAMS>javac Matrix.java

D:\Documents\Java\LAB PROGRAMS>java Matrix
Enter the number of rows: 3
Enter the number of coloumns in row 1
2
Enter the number of coloumns in row 2
3
Enter the number of coloumns in row 3
4
Enter the elements of row 1
1 2
Enter the elements of row 2
3 4 5
Enter the elements of row 3
6 7 8 9

The given Matrix
1 2
3 4 5
6 7 8 9

Sum of row 1 = 3
Sum of row 2 = 12
Sum of row 3 = 30
```

7. Program to extract portion of character string and print extracted string. Assume that 'n' characters extracted starting from mth character position.

Code:

```
import java.util.*;

class ExtractString {

    public static void main(String args[]) {

        Scanner sc = new Scanner(System.in);

        String str1, str2;

        int m = 0, n = 0;

        System.out.print("Enter a String: ");

        str1 = sc.nextLine();

        System.out.print("Enter numbers of characters to be extracted from string:
");

        m = sc.nextInt();

        System.out.print("Enter Starting Index: ");

        n = sc.nextInt();

        System.out.println("The given string is: " + str1);

        str2 = str1.substring(n, (m + n));

        System.out.println("The extracted from string is: " + str2);

    }

}
```

Output:

```
Microsoft Windows [Version 10.0.26200.8037]
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D:\Documents\Java\LAB PROGRAMS>javac ExtractString.java

D:\Documents\Java\LAB PROGRAMS>java ExtractString
Enter a String: Ashwith Frank
Enter numbers of characters to be extracted from string: 7
Enter Starting Index: 0
The given string is: Ashwith Frank
The extracted from string is: Ashwith
```

8. Program to add, remove and display elements of a Vector.

Code:

```
import java.util.Vector;
import java.util.Scanner;
public class VectorList {
    public static void main(String args[]) {
        Vector v = new Vector(6, 2);
        String item;
        int ch = 0, i = 0, n;
        Scanner sc = new Scanner(System.in);
        while (ch != 5) {
            System.out.println("\n1.Append \n2.Insert \n3.Delete \n4.Display
\n5.Exit");
            System.out.print("Enter your choice: ");
            ch = sc.nextInt();
            switch (ch) {
                case 1:
                    System.out.print("Enter the item to append: ");
                    item = sc.next();
                    v.add(item);
                    break;
                case 2:
                    System.out.print("Enter the item to insert: ");
                    item = sc.next();
                    System.out.print("Enter the index: ");
                    i = sc.nextInt();
```

```
                v.insertElementAt(item, i);
                break;
            case 3:
                System.out.print("Enter the item to delete: ");
                item = sc.next();
                if (v.remove(item) == true)
                    System.out.println(item + " deleted from the list");
                else
                    System.out.println(item + " not found in the list");
                break;
            case 4:
                n = v.size();
                for (i = 0; i < n; i++) {
                    item = (String) v.elementAt(i);
                    System.out.println(item);
                }
        }
    }
  }
}
```

Output:

```
Microsoft Windows [Version 10.0.26200.8037]
(c) Microsoft Corporation. All rights reserved.

D:\Documents\Java\LAB PROGRAMS>javac VectorList.java
Note: VectorList.java uses unchecked or unsafe operations.
Note: Recompile with -Xlint:unchecked for details.

D:\Documents\Java\LAB PROGRAMS>java VectorList

1.Append
2.Insert
3.Delete
4.Display
5.Exit
Enter your choice: 1
Enter the item to append: Apple

1.Append
2.Insert
3.Delete
4.Display
5.Exit
Enter your choice: 1
Enter the item to append: Banana

1.Append
2.Insert
3.Delete
4.Display
5.Exit
Enter your choice: 2
Enter the item to insert: Mango
Enter the index: 1

1.Append
2.Insert
3.Delete
4.Display
5.Exit
Enter your choice: 3
Enter the item to delete: Apple
Apple deleted from the list

1.Append
2.Insert
3.Delete
4.Display
5.Exit
Enter your choice: 4
Mango
Banana

1.Append
2.Insert
3.Delete
4.Display
5.Exit
Enter your choice: 5
```